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Polymarket Bias & Mispricing Analysis

Closed Markets 2025 — Calibration & Category Mispricings

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2026-02-18

1 · Setup & Data Loading

Show

Show

```
## 111178 market rows loaded
```

Show

```
## 87 columns
```

1.1 Parse Resolutions & Token Mapping

We need to know (a) which token is the “Yes” token and (b) whether that outcome resolved to 1.

Show

```
## 49753 binary Yes/No markets with clean resolution
```

Show

```
## Resolution rate (Yes = 1): 0.267
```

1.2 Derive Categories from Tags

Show

```
##
## ── Category distribution ──
```

Show

```
## # A tibble: 8 × 2
##   category          n
##   <chr>         <int>
## 1 Sports        14881
## 2 Crypto        14161
## 3 Politics       9844
## 4 Science/Tech   5570
## 5 Other          2410
## 6 Entertainment  2091
## 7 Economics       777
## 8 World Events     19
```

2 · Calibration Analysis (Snapshot Prices)

The core question: **When the market prices a Yes token at probability p, does the event actually happen p% of the time?**

We use the price history to sample market prices at fixed intervals before resolution, then check calibration.

Show

```
## 49753 unique Yes tokens to look up
```

Show

```
## 100 batches of up to 500 tokens
```

Show

```
##   Batch 5/100 done (11595 snapshot rows so far)
##   Batch 10/100 done (23809 snapshot rows so far)
##   Batch 15/100 done (36470 snapshot rows so far)
##   Batch 20/100 done (48962 snapshot rows so far)
##   Batch 25/100 done (61591 snapshot rows so far)
##   Batch 30/100 done (74504 snapshot rows so far)
##   Batch 35/100 done (87249 snapshot rows so far)
##   Batch 40/100 done (100113 snapshot rows so far)
##   Batch 45/100 done (112551 snapshot rows so far)
##   Batch 50/100 done (123420 snapshot rows so far)
##   Batch 55/100 done (133622 snapshot rows so far)
##   Batch 60/100 done (145422 snapshot rows so far)
##   Batch 65/100 done (157735 snapshot rows so far)
##   Batch 70/100 done (169985 snapshot rows so far)
##   Batch 75/100 done (181988 snapshot rows so far)
##   Batch 80/100 done (194828 snapshot rows so far)
##   Batch 85/100 done (207703 snapshot rows so far)
##   Batch 90/100 done (220378 snapshot rows so far)
##   Batch 95/100 done (232904 snapshot rows so far)
##   Batch 100/100 done (245827 snapshot rows so far)
```

Show

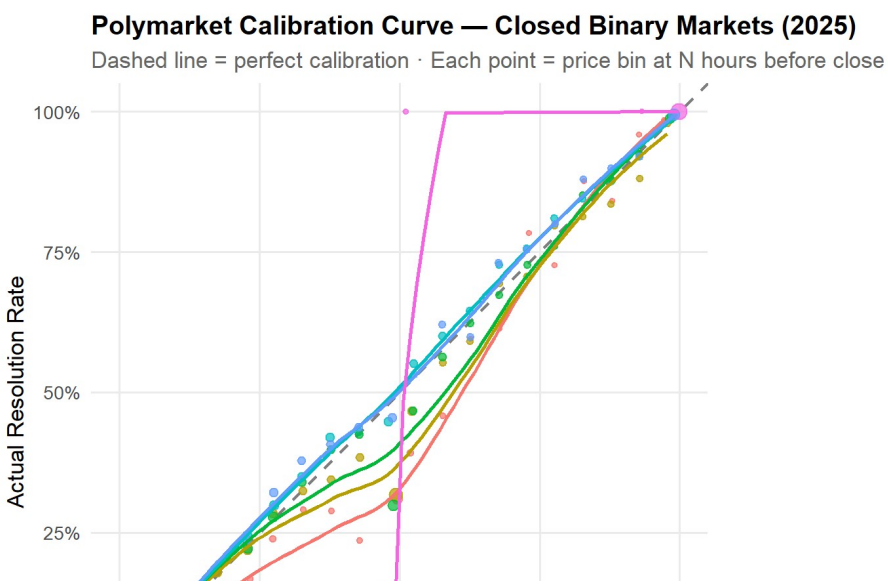
```
##            used  (Mb) gc trigger  (Mb)  max used  (Mb)
## Ncells  3499346 186.9    5750472 307.2   5750472 307.2
## Vcells 42131714 321.5  121330323 925.7 121329831 925.7
```

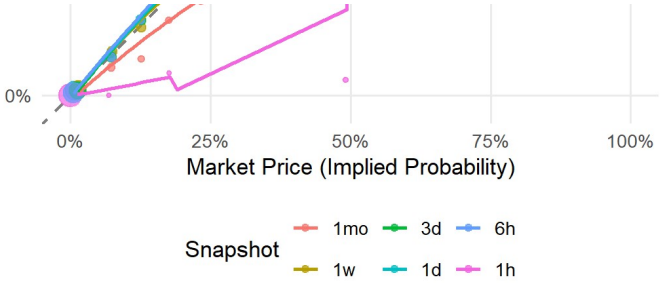
Show

```
##
##  245827 total market × snapshot observations
```

2.1 Overall Calibration Curve

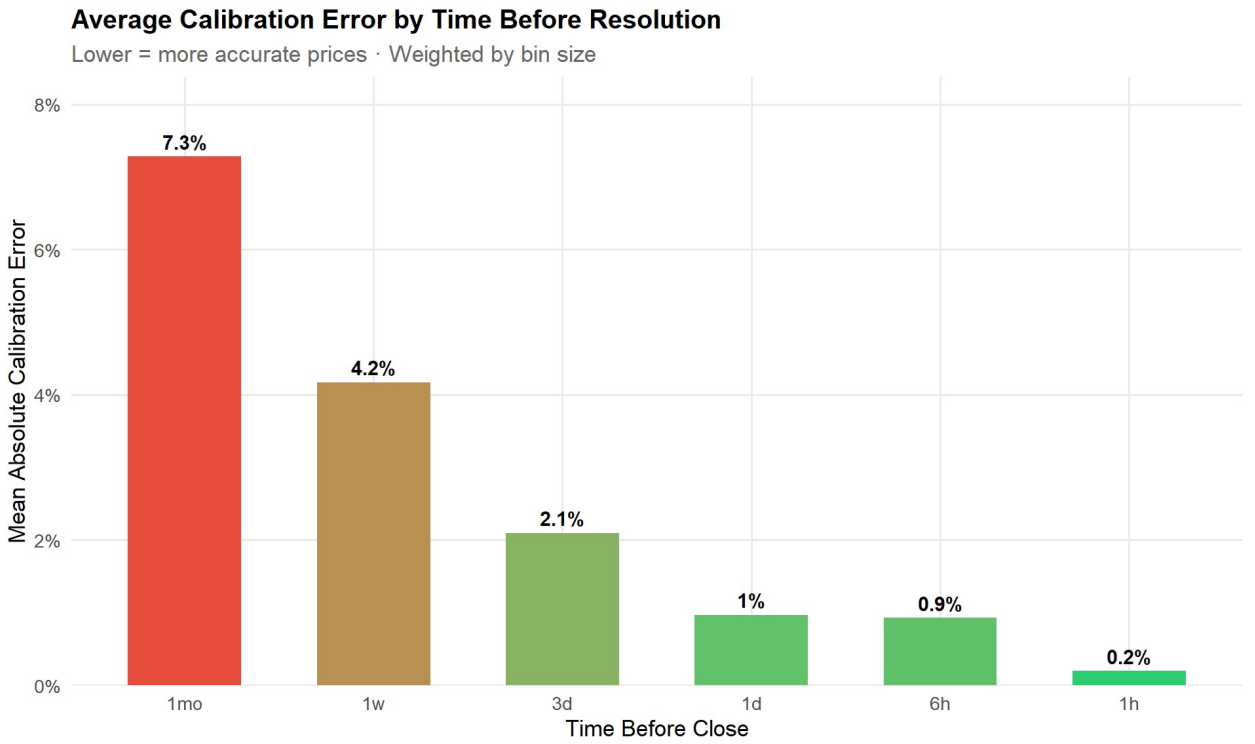
Show





2.2 Calibration Error by Snapshot Horizon

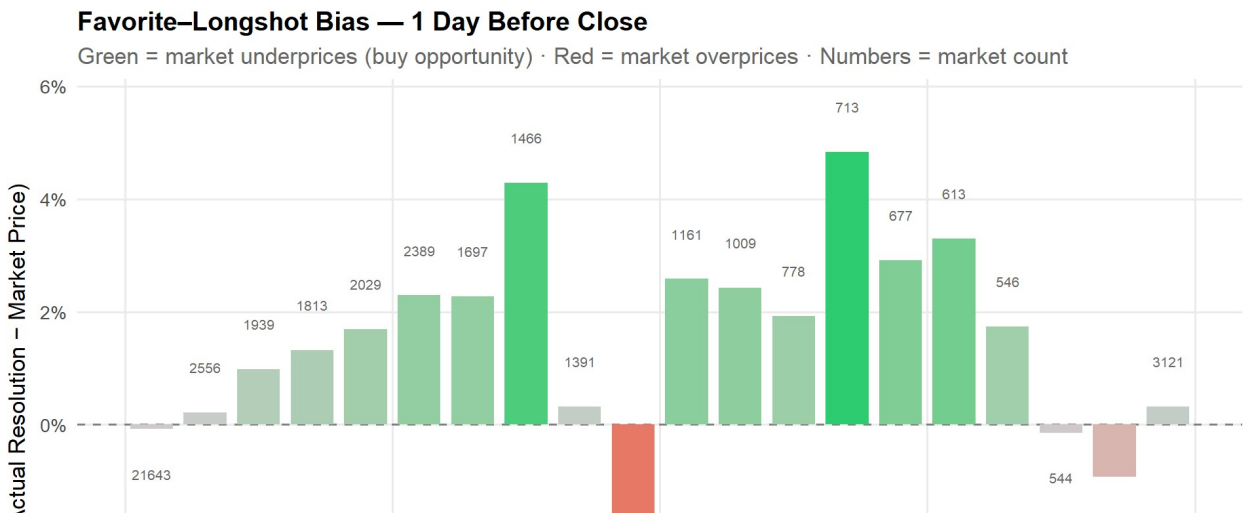
Show

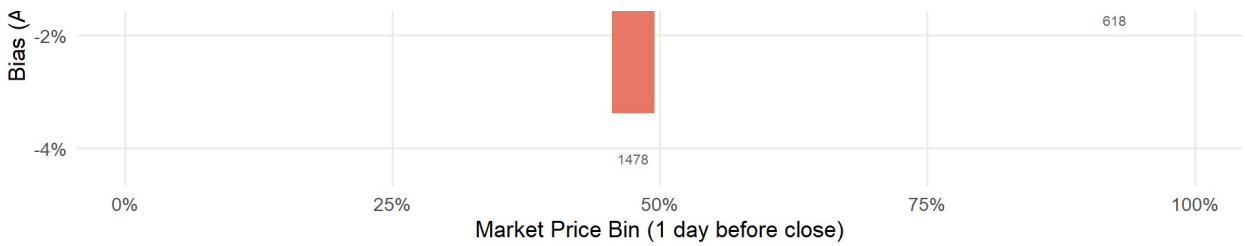


2.3 Favorite–Longshot Bias (within Calibration)

Even if overall calibration looks good, a **favorite–longshot bias** means longshots (low probability) are overpriced and favorites (high probability) are underpriced — or vice versa.

Show

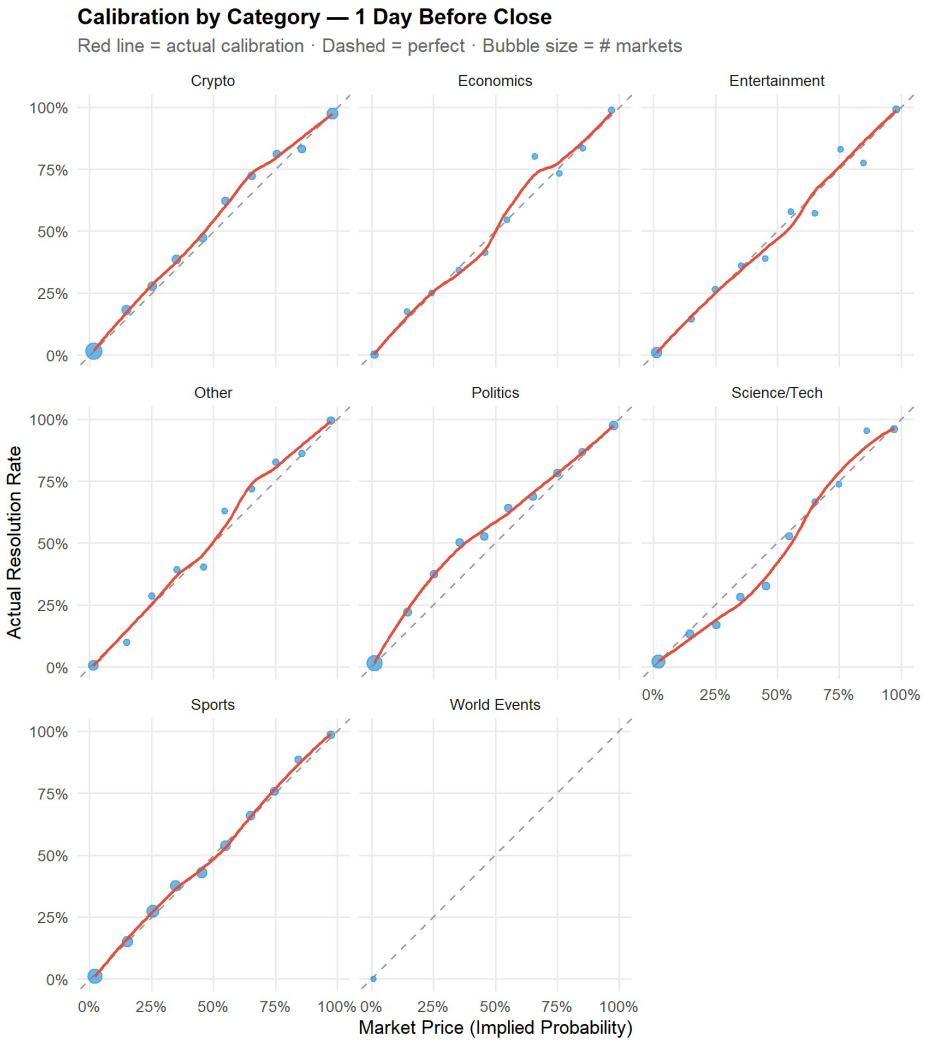




3 · Category-Level Mispricings

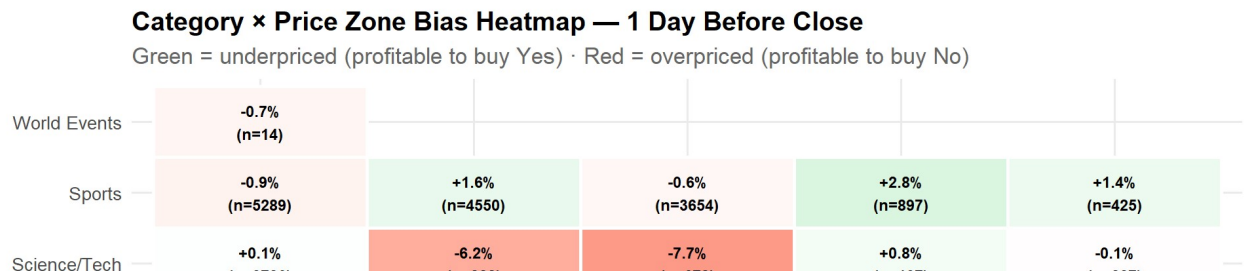
3.1 Category Calibration Curves

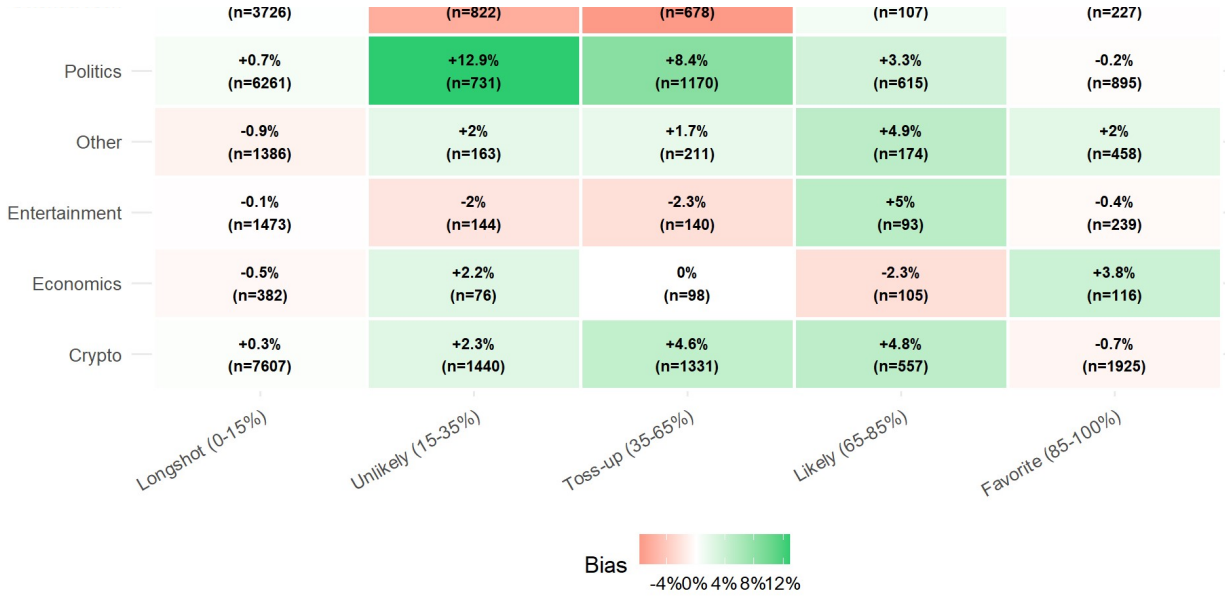
Show



3.2 Category Bias Heatmap

Show





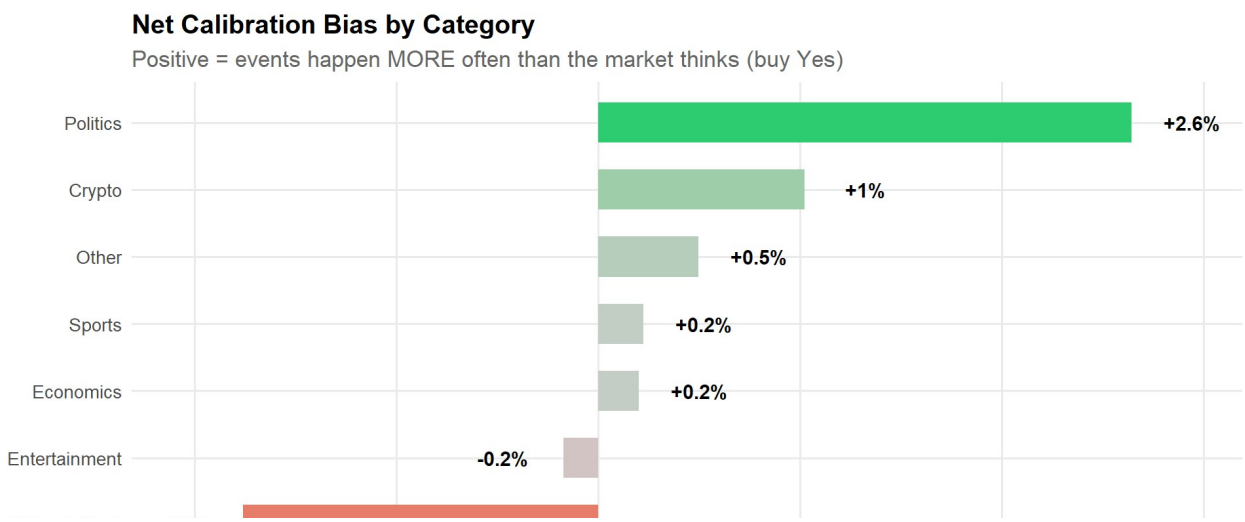
3.3 Aggregate Category Bias & Brier Score

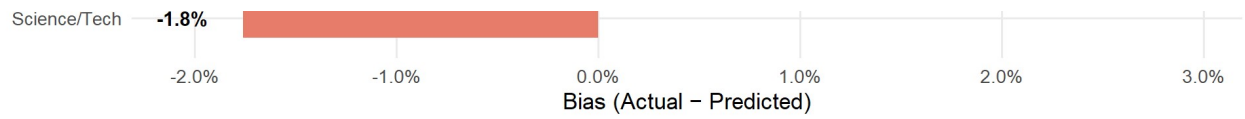
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Category-Level Bias & Accuracy (1 day before close)

category	n_markets	avg_price	resolution_rate	bias	brier_score	abs_bias
World Events	18	0.1137	0.2222	0.1085	0.0650	0.1193
Politics	9672	0.2265	0.2529	0.0264	0.0755	0.1431
Science/Tech	5560	0.1656	0.1480	-0.0176	0.0743	0.1533
Crypto	12860	0.2694	0.2796	0.0102	0.0732	0.1438
Other	2392	0.3098	0.3148	0.0050	0.0554	0.1200
Sports	14815	0.2823	0.2846	0.0022	0.1407	0.2826
Economics	777	0.3403	0.3423	0.0020	0.0846	0.1735
Entertainment	2089	0.2061	0.2044	-0.0017	0.0473	0.0972

Show

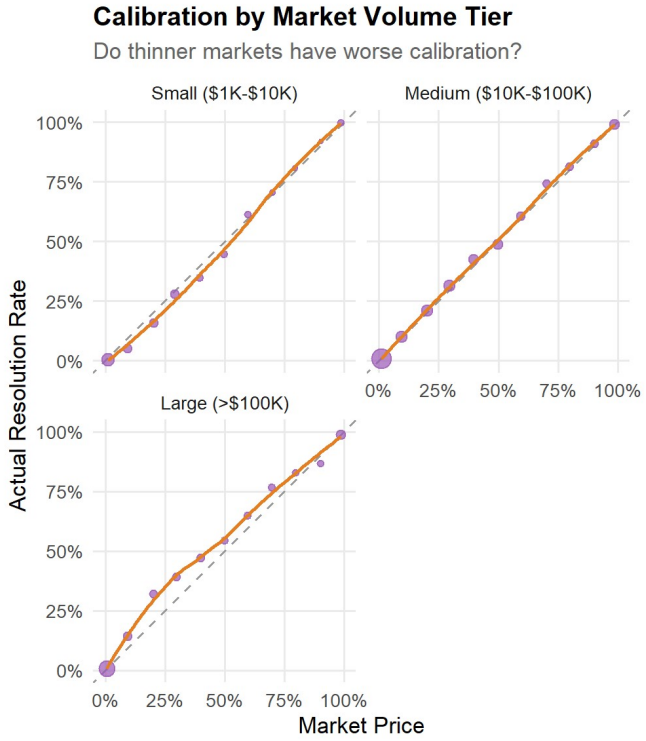




4 · Volume-Weighted Analysis

Thin markets are noisier. Let's check if bias patterns change when we weight by volume.

Show



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Brier Score & Bias by Volume Tier

vol_tier	n	brier_score	bias
Small ($1K-$10K)	8650	0.0917	-0.0188
Medium ($10K-$100K)	26440	0.0958	0.0058
Large (>$100K)	13093	0.0872	0.0263

5 · Exploitable Pattern Summary

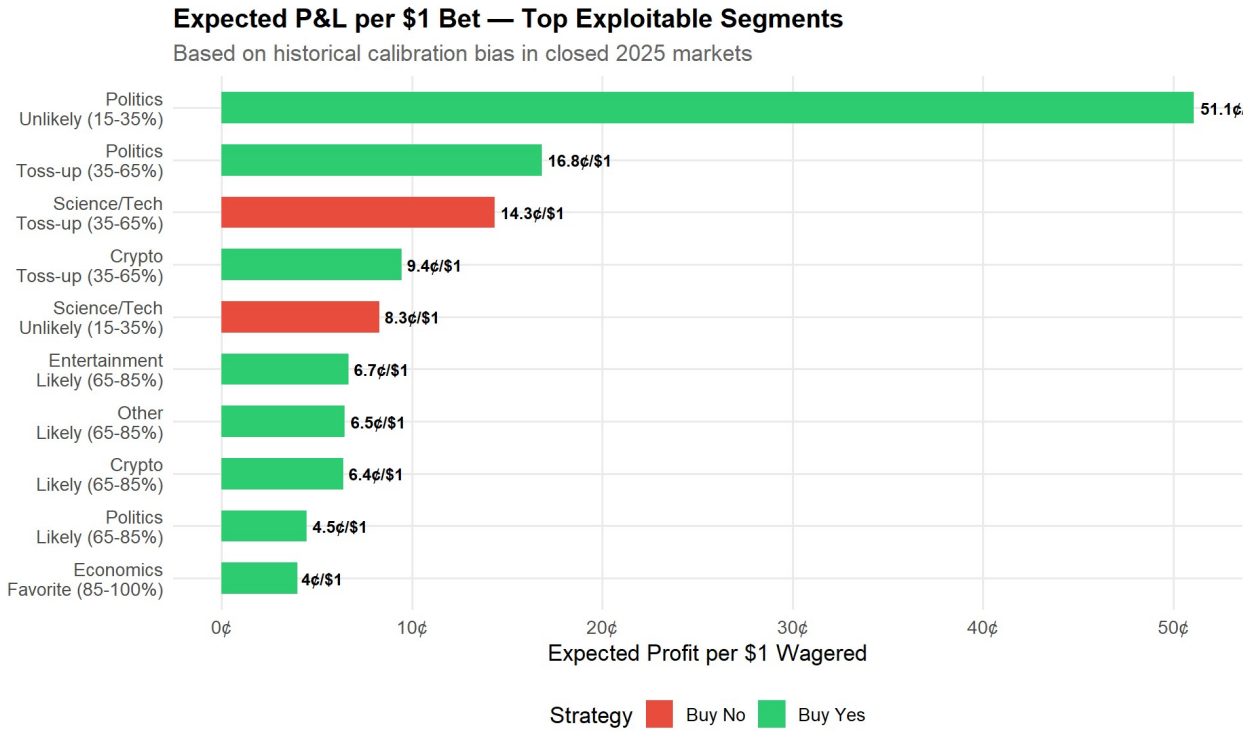
Show

Top 20 Most Exploitable Category × Price Zone Combinations (|bias| > 3%, n ≥ 15)

category	price_zone	n_markets	avg_price	actual_rate	bias	edge_pct	brier
Politics	Unlikely (15-35%)	731	0.253	0.382	0.129	51.071	0.247

category	price_zone	n_markets	avg_price	actual_rate	bias	edge_pct	brier
Politics	Toss-up (35-65%)	1170	0.502	0.586	0.084	16.802	0.247
Science/Tech	Toss-up (35-65%)	678	0.465	0.388	-0.077	-16.506	0.232
Science/Tech	Unlikely (15-35%)	822	0.251	0.189	-0.062	-24.751	0.155
Entertainment	Likely (65-85%)	93	0.756	0.806	0.050	6.656	0.152
Other	Likely (65-85%)	174	0.761	0.810	0.049	6.471	0.156
Crypto	Likely (65-85%)	557	0.751	0.799	0.048	6.371	0.164
Crypto	Toss-up (35-65%)	1331	0.482	0.527	0.046	9.444	0.243
Economics	Favorite (85-100%)	116	0.945	0.983	0.038	3.976	0.019
Politics	Likely (65-85%)	615	0.749	0.782	0.033	4.470	0.169

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6 · Key Takeaways

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Automated Findings

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- **Overall market bias**: +0.7% (1 day before close). Markets are approximately well-calibrated overall.

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- **Most biased category**: World Events (+10.8% bias, n=18)

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- **Best calibrated category**: Entertainment (Brier score: 0.0473)

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- **Largest exploitable edge**: Politics × Unlikely (15-35%) (+12.9% bias, n=731)

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> **Disclaimer**: Past calibration patterns do not guarantee future exploitability. Transaction costs, slippage, and market evolution may erode these edges.

Analysis generated on 2026-02-18 using 49753 closed binary markets and price history data.